

Analysis of 3000 T class submarines equipped with polymer electrolyte fuel cells

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ABSTRACT

The naval submarines have conventionally been equipped with diesel-electric propulsion. The diesel generators charge the batteries when the submarine is at the surface or at snorkelling depth. This is the biggest short-coming of this system as the submarine can be detected due to the infrared signatures from the exhaust of engines. Present study aims in analysing the feasibility of using fuel cells as a replacement of conventional diesel based system. Fuel cell system is analysed to meet the propulsion load and hotel load. In this purpose, metal hydride and sodium borohydride are considered for fuel and compressed oxygen and liquid oxygen are considered as oxidant. The most effective combination with respect to weight, volume has been analysed. The submerged endurance and distance for various hotel loads under submerged conditions have also been estimated. It is found that the metal hydride and liquid oxygen combination can be easily retrofitted by replacing the conventional system. However, MH/O_2 , SBH/O_2 and SBH/LO_x require some extra room to be created. All the systems show substantial enhancement in the submerged endurance.

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1. Introduction

Submarines are valuable assets to any Navy due to their inherent capability of sailing while submerged under water and launching attacks unnoticed. This makes its detection difficult and makes patrolling in hostile water at the time of conflict safer than ships. At present most of the submarines are diesel engine/battery propelled. However, the conventional diesel-electric submarines have their own limitations. The submerged time is limited by the capacity of batteries and they are prone to detection when surfaced for charging batteries or at few meters below the surface known as snorkelling depth. In addition to the strong infrared signature emitted by the exhausts of the diesel generators during snorkelling, the boats are susceptible to visual detection.

In order to enhance the submerged time of the existing submarines various options are available, aptly called 'Air Independent Propulsion' options as they facilitate the submarine to be submerged for prolonged duration by avoiding the need to surface for air access. These include Closed Cycle Diesel (CCD), Sterling engine, Module d'Energie Sous-Marin Autonome (MESMA), Closed Cycle Gas Turbine (CCGT) and low temperature fuel cells.

For low temperature operations, Polymer Electrolyte Fuel Cells (PEFCs) are the most favourable [1–18] option for submarines.

Hydrogen for the reaction can mainly be stored onboard in the form of compressed gas [19,20], in liquid form at cryogenic temperatures [21] or in solid form in some hydrides [22,23]. Studies comparing all storage options [24–27] favour the storing of hydrogen and oxygen or air in gaseous state. However, for submarine applications compressed gases may not be an answer due to safety and reliability constraints. Sodium borohydride (SBH) can be considered as a hydrogen carrier [28–33] which might give higher weight percent of hydrogen storage than compressed hydrogen. Liquid oxygen (LO_x) is considered as the oxidant. Compressed air which is carried onboard in present system for other purposes has also been considered as a variant oxidant.

In the present work, a feasibility study has been carried out to equip a generic 3000 ton submarine by replacing the diesel based conventional system with PEFC and battery to work out the optimum combination, considering weight and volume, towards Air Independent Propulsion (AIP). Finally, the duration in submerged condition is being investigated for different combinations and compared with conventional system.

2. System description

Conventional diesel based electric submarines mainly comprise of diesel engines, battery banks, motors used for sprinting at high speed (14–16 knots; 1 knot $\approx 1.852 \text{ km h}^{-1}$) and sailing at low speed (4–5 knots). The diesel generators are mainly used to charge the batteries only when the submarine has the access to the

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Nomenclature

D	submerged displacement [kg]
E_{vol}	volumetric energy density of hydrogen [kW h m^{-3}]
E_{BAT}	energy supplied by battery bank [kW h]
E_{hotel}	daily energy demand by hotel load [kW h d^{-1}]
E_T	daily energy demand [kW h d^{-1}]
E_{hor}	energy required for leaving and entering harbour [kW h]
E_{store}^W	total stored energy in the submarine based on available weight [kW h]
E_{store}^V	total stored energy in the submarine based on available volume [kW h]
M_{H_2}	molecular weight of hydrogen [g mol^{-1}]
N_{H_2}	number of moles of hydrogen
N_{O_2}	number of moles of oxygen
P_{AV}	average power required from fuel cell [kW]
P_{FC}	fuel cell stack size [kW]
P_{eco}	propulsion power at economic speed [kW]
P_{prop}	propulsion load [kW]
P_{sprint}	propulsion power at high speed [kW]
P_{hotel}	hotel load [kW]
P_{tank}	oxygen tank pressure [MPa]
$Q_{batbank}$	battery bank capacity [kW h]
Q_{PERC}	battery storage capacity [%]
S	total distance travel by submarine [km]
T_{end}^W	endurance based on weight [d]
T_{end}^V	endurance based on volume [d]
T_{DIS}	ratio of discharge rate
T_{NOM}	nominal discharge [h]
T_{ACT}	actual discharge [h]
T	time of battery operation per day [h d^{-1}]
T_{eco}	total propulsion time at low speed [h]
T_{sprint}	total propulsion time at high speed [h]
V_{DGBAT}	available volume in submarine [m^3]
V_{SBH}	volume of sodium borohydride [$\text{m}^3 \text{d}^{-1}$]
V_{H_2O}	volumes of water [$\text{m}^3 \text{d}^{-1}$]
V_{BOP}	volume of balance of plant [m^3]

V_{H_2}	volume of hydrogen at STP [$\text{m}^3 \text{d}^{-1}$]
V_{O_2SYS}	volume of the compressed oxygen storage system [$\text{m}^3 \text{d}^{-1}$]
V_{LOxSYS}	volume of the liquid oxygen storage system [$\text{m}^3 \text{d}^{-1}$]
V_{H_2SYS}	total volume of hydrogen storage system [$\text{m}^3 \text{d}^{-1}$]
V_{SBHSYS}	total volume of sodium borohydride storage system [$\text{m}^3 \text{d}^{-1}$]
$V_{fuelsys}$	total volume of fuel system [$\text{m}^3 \text{d}^{-1}$]
V_{oxisys}	total volume of oxidant system [$\text{m}^3 \text{d}^{-1}$]
$W_{fuelsys}$	total weight of fuel system [kg d^{-1}]
W_{oxisys}	total weight of oxidant system [kg d^{-1}]
W_{SBH}	weight of sodium borohydride [kg d^{-1}]
W_{H_2O}	weights water [kg d^{-1}]
W_{BOP}	weight of balance of plant [kg]
WT_{perc}	gravimetric storage in wt%
W_{DGBAT}	available weight in submarine [kg]
W_{LOxSYS}	weight of liquid oxygen system [kg d^{-1}]
W_{O_2}	weight of oxygen [kg d^{-1}]
W_{H_2}	weight of hydrogen [kg d^{-1}]
W_{H_2SYS}	weight of hydrogen storage [kg d^{-1}]
W_{SBHSYS}	total weight of sodium borohydride storage system [kg d^{-1}]
p_{gra_FC}	gravimetric power density of fuel cell stacks [kW kg^{-1}]
p_{vol_FC}	volumetric power density of fuel cell stacks [kW m^{-3}]
r_{grav}	gravimetric fraction LOx
r_{vol}	volumetric fraction LOx
u	speed of the submarine [km h^{-1}]
v_{eco}	propulsion power at low speed [km h^{-1}]
v_{sprint}	propulsion power at high speed [km h^{-1}]

Greek symbols

ΔH	higher heating value [kW h mol^{-1}]
η_{DOD}	depth of discharge of the battery bank
$\eta_{OVERALL}$	overall efficiency
η_B	battery efficiency
$\eta_{DC/DC}$	efficiency of DC/DC converter
η_{FC}	efficiency of fuel cell
ρ_{LOx}	density of liquid oxygen [kg m^{-3}]

atmosphere, enabling combustion. The batteries fulfill the total energy requirements of the submarine during submerged condition. The schematic diagram of a conventional submarine is shown in Fig. 1.

The total load requirement of the submarine is categorised mainly as hotel load and propulsion load. The propulsion load of the submarine is a function of the speed (u) and the submerged displacement (D) which is given by [5],

$$P_{prop} = 0.0026 \cdot \left(\frac{D}{1000}\right)^{2/3} \cdot \left(\frac{u}{1.852}\right)^3 \quad (1)$$

The peak power requirement is mainly controlled by the maximum attainable speed of the submarine.

In addition to the propulsion load, the submarine has a large set of equipments, which include pumps and blowers, communication sets, life support systems, domestic appliances, control systems and armament. The hotel loads generally remain constant for specific requirements and it may vary between 100 kW and 475 kW based on the requirements.

However, the operating duration and total energy consumptions by different equipments vary widely. To estimate the load profile for the submarine, it is assumed that the submarine sails at a speed of 5 knots using low speed motor for 22 h and sprints using high-

speed motor for 2 h at 16 knots each day. During two hours of sprinting, it requires almost 5805 kW h. However, two hours of sprinting time is subject to variation depending upon practical speed requirements and situation demands. The assumed typical daily load profile is shown in Fig. 2.

The load profile shows three distinct levels of operation. High-speed motor and low speed motor cover the propulsion load while pumps cover all pumps used onboard. The miscellaneous head includes the energy consumed during entering and leaving harbour, air compressors and domestic load. The other heads include load of air conditioning and refrigeration system besides alternator loads.

The total weight and volume of the system used for power generation and energy storage is important as it is planned in the

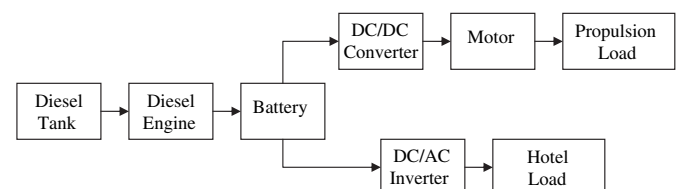


Fig. 1. Schematic diagram for the conventional system.

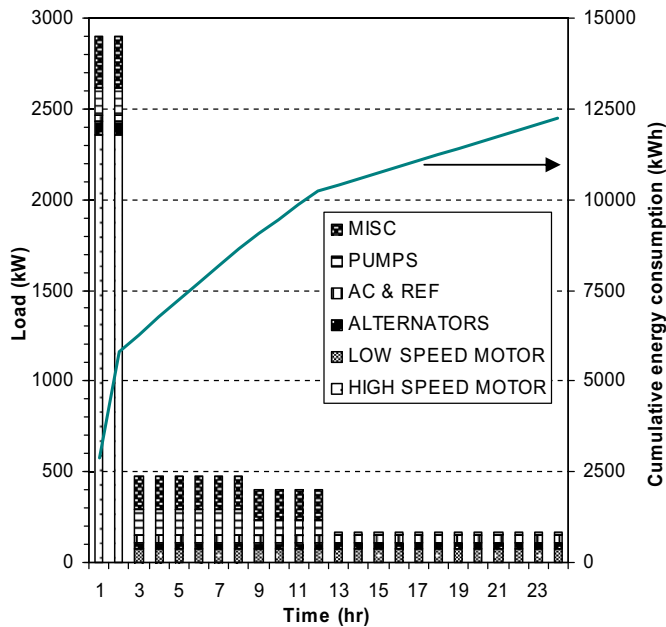


Fig. 2. Daily load profile combining propulsion and hotel load.

present work to replace the conventional power system with a fuel cell system. The total volume and weight for different components in the conventional system could be replaced are given in Table 1.

As an alternate to the conventional diesel-electric system a fuel cell and battery-based system is proposed and different combinations are being evaluated. In this system, hydrogen and oxygen will be carried to power the fuel cell, which will fulfill the energy requirements of the submarine. The schematic diagram of the fuel cell/battery-based system is shown in Fig. 3.

In the present study, three different types of systems combinations are being considered such as only fuel cell based system, only battery-based system and fuel cell and battery-based hybrid system as explained below:

- (i) *FC based system*: represents a solely fuel cell based system where the fuel cells replace the engines and batteries of conventional system, hydrogen system and oxygen storage come in lieu of diesel and lubricating oil.
- (ii) *FC-Bat hybrid*: stands for the fuel cell-battery-based hybrid system where battery provides the energy while sprinting and fuel cells provide power for rest of the twenty two hours which includes charging the batteries. Hence, the fuel cell size can be reduced substantially.
- (iii) *Battery-based system*: is independent of fuel cells. Thus, the submarine for such a case has to charge the batteries completely before sailing and the endurance at sea is dependent only upon the one time battery charge.

In the proposed system, PEMFCs are used and combinations of battery and fuel cells are considered for fulfilling the energy

Table 1
Approximate weight and volume of replaceable components of conventional system.

Components	Number	Weight (kg)	Volume (m ³)
Engine	2	30,000	34
Fuel	1	172,000	212
Lubricant	1	12,000	13
Battery	240	192,000	68
Total		406,000	327

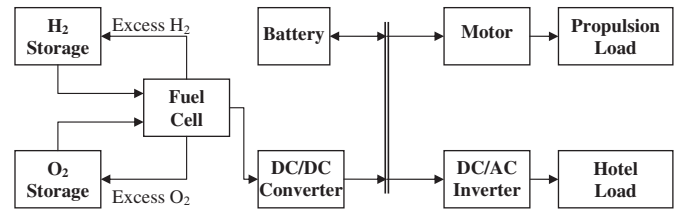


Fig. 3. Schematic diagram for the fuel cell/battery-based system.

requirement. Metal hydride for hydrogen storage and sodium borohydride as hydrogen carrier is being explored. For oxidant, liquid oxygen and compressed oxygen at 70 MPa are being considered. The requirement to carry diesel fuel and lubricating oil is eliminated and the space available is used for carrying the fuel cell systems and fuels.

3. System modelling

The daily load distribution shown in Fig. 2 has been utilised for sizing the components of the fuel cell based system. The different equipments are utilised for different durations in a day and can be categorised into distinct types based on total instantaneous load demand. For the model formulation, following assumptions are made for the proposed system:

- The load has been categorised according to hours of usage each day.
- The propulsion load is assumed to be divided between high speed (sprinting) at 16 knots and low speed (sailing) at 5 knots for 2 and 22 h respectively in a day.
- Excess gases (H₂, O₂) from fuel cell stack are recycled into the system.
- During battery operation, in hybrid system, the full energy demand is supplied by the battery and fuel cells are not operated.
- Fuel cells supply power for load as well as charging batteries simultaneously.
- The efficiencies for DC/DC converter, motor, battery are assumed to be constant.

3.1. Battery bank

The available capacity of the battery bank depends upon the rate at which the battery is discharged. As two different combinations of battery with fuel cell are considered in present study, the discharge duration of the battery varies are 2 h and 24 h for hybrid system and only battery system respectively. Hence, the discharge rate varies almost 12 folds causing a substantial variation in the available battery capacity. As the discharge durations of battery in the different systems are different, it has been normalised using the ratio of discharge rates as follows and the available capacity for the given battery size is estimated:

$$T_{DIS} = \frac{T_{NOM}}{T_{ACT}} \quad (2)$$

where T_{DIS} is ratio of discharge rate, T_{NOM} is nominal discharge (h) duration and T_{ACT} is actual discharge (h) duration.

The storage capacity of the batteries decreases as the discharge rate is increased. For present study, the nominal discharge rate is considered to be 20 h which indicates that the ratio of discharge rate lies between 0.8 and 10.

The variation of actual available storage capacity with the discharge rate is shown in Fig. 4 and the co-relation between the

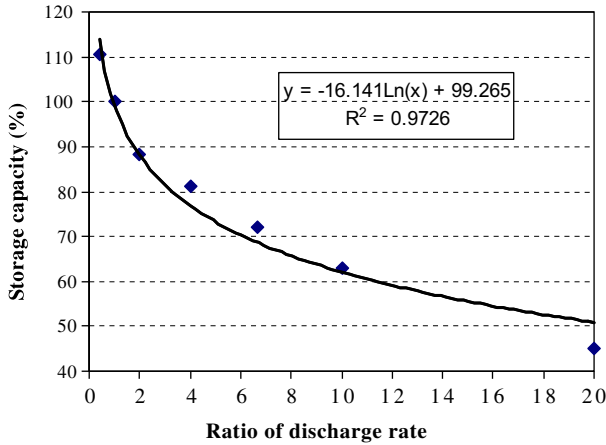


Fig. 4. Variation in the storage capacity with the discharge rate [34].

discharge ratio and actual available storage capacity for the battery used in conventional system is given by,

$$Q_{\text{PERC}} = 99.265 - 16.141 \ln(T_{\text{DIS}}) \quad (3)$$

where Q_{PERC} is the storage capacity in percent.

Q_{PERC} is used to estimate the total battery bank capacity required to fulfill the energy requirements which is given by,

$$Q_{\text{batbank}} = E_{\text{BAT}} \cdot \frac{100}{Q_{\text{PERC}}} \cdot \frac{1}{\eta_{\text{DOD}}} \quad (4)$$

where η_{DOD} is depth of discharge of the battery bank and E_{BAT} is total energy supplied by battery bank in kWh.

3.2. Fuel cells

Fuel cells are being sized comparing the peak load and average load for only FC system and hybrid system respectively. The average load can be calculated as,

$$P_{\text{AV}} = \frac{E_T}{(24 - T)} \quad (5)$$

where E_T is daily energy demand in kWh, P_{AV} is average power required from fuel cell in kW and T is time (h) of battery operation per day.

3.3. Fuel storage system

Generally, pure hydrogen is considered as the fuel for PEFC applications which can be stored in metal hydride form; however, such system has very low volumetric and gravimetric energy densities. As a result of that it requires comparatively larger space and weight. Chemical hydrides are very attractive options as hydrogen carrier to supply hydrogen at room temperature. Among various options, aqueous sodium borohydride (NaBH_4) seems to be a perfect hydrogen source as it is stable, non-flammable and non-toxic in nature with hydrogen storing capability of 10.66 wt%. The borax, obtained as a product after removal of hydrogen from NaBH_4 , is environmentally clean and can be recycled to generate the NaBH_4 . Use of sodium borohydride as hydrogen carrier improves the gravimetric energy density and volumetric energy densities.

However, for the submarine in the present approach the use of fuel cell based systems depends on available volume and weight of the system after replacing the conventional diesel-battery-based

hybrid system. Irrespective of the storage system, the total amount of hydrogen in terms of moles required for meeting the total energy demand (E_T) is given by,

$$N_{\text{H}_2} = \frac{E_T}{\Delta H} \cdot \frac{1}{\eta_{\text{OVERALL}}} \quad (6)$$

where ΔH is the higher heating value in kWh mol^{-1} , η_{OVERALL} is overall efficiency of the system and is given by,

$$\eta_{\text{OVERALL}} = \eta_B \cdot \eta_{\text{DC/DC}} \cdot \eta_{\text{FC}} \quad (7)$$

where η_B is efficiency of battery, $\eta_{\text{DC/DC}}$ is efficiency of DC/DC converter and η_{FC} is efficiency of fuel cell.

For present calculation the HHV is considered because PEFC always operates in the ranges $<100^\circ\text{C}$.

Total weight of stored hydrogen in kg is given by,

$$W_{\text{H}_2} = N_{\text{H}_2} \cdot M_{\text{H}_2} \cdot 10^{-3} \quad (8)$$

where M_{H_2} is molecular weight of hydrogen (g mol^{-1}).

Hence, the total weight of the hydrogen storage system is given by,

$$W_{\text{H}_2\text{SYS}} = \frac{W_{\text{H}_2} \cdot 100}{W_{\text{Tperc}}} \quad (9)$$

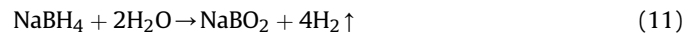
where W_{Tperc} is gravimetric storage in percent.

The total volume of hydrogen storage system (m^3) is given by,

$$V_{\text{H}_2\text{SYS}} = \frac{E_T}{E_{\text{vol}}} \cdot 10^{-3} \cdot \frac{1}{\eta_{\text{OVERALL}}} \quad (10)$$

where E_{vol} is volumetric energy density in kWh L^{-1} .

In case of NaBH_4 as hydrogen carrier, hydrogen is generated by following hydrolysis reaction:



Hence, for manufacturing hydrogen from NaBH_4 water has to be provided. However, it is very interesting to note that water also acts as hydrogen source.

The total weight of NaBH_4 system is given by following equations:

$$W_{\text{SBHSYS}} = W_{\text{SBH}} + W_{\text{H}_2\text{O}} + W_{\text{BOP}} \quad (12)$$

where W_{BOP} is the weight of balance of plant, W_{SBH} and $W_{\text{H}_2\text{O}}$ are the weights of NaBH_4 and water which can be estimated from the following equations:

$$W_{\text{SBH}} = N_{\text{H}_2} \cdot \frac{M_{\text{SBH}}}{4} \cdot 10^{-3} \quad (13)$$

where M_{H_2} is molecular weight of SBH (g mol^{-1}).

$$W_{\text{H}_2\text{O}} = N_{\text{H}_2} \cdot \frac{2 \cdot M_{\text{H}_2\text{O}}}{4} \cdot 10^{-3} \quad (14)$$

The total volume of NaBH_4 system is given by following equations:

$$V_{\text{SBHSYS}} = V_{\text{SBH}} + V_{\text{H}_2\text{O}} + V_{\text{BOP}} \quad (15)$$

where V_{BOP} is the volume of balance of plant, V_{SBH} and $V_{\text{H}_2\text{O}}$ are the volumes of NaBH_4 , and water which can be estimated from the following equations:

$$V_{\text{SBH}} = \frac{W_{\text{SBH}}}{\rho_{\text{SBH}}} \quad (16)$$

where ρ_{SBH} is the density of SBH in kg m^{-3} .

$$V_{H_2O} = \frac{W_{H_2O}}{\rho_{H_2O}} \quad (17)$$

where ρ_{H_2O} is the density of water in kg m^{-3} .

3.4. Oxidant storage system

Oxygen is being considered as oxidant for PEFC. However, air can be also considered as oxidant for PEFC. The main issue with air is related to the presence of lower amount of oxygen leading to a bulky storage system. As for PEFC reactions one mole of hydrogen reacts with half mole of oxygen occupies exactly half of the hydrogen volume. However, when air is used as oxidant it occupies almost 2.5 times of hydrogen volume. Hence, in the present investigation air is not considered as the source of oxygen. For the present investigation compressed oxygen at 70 MPa and liquid oxygen are considered.

The total volume of oxygen system is computed with respect to volume of hydrogen system and is half of it. The number of mole and weight of oxygen required is given by

$$N_{O_2} = \frac{N_{H_2}}{2} \quad (18)$$

$$W_{O_2} = 8 \cdot W_{H_2} \quad (19)$$

The weight of compressed and liquid oxygen storage system can be calculated as follows,

$$W_{LO_2SYS/O_2SYS} = \frac{N_{O_2} M_{O_2}}{r_{\text{grav}}} 10^{-3} \quad (20)$$

where r_{grav} is the gravimetric fraction and M_{O_2} is the molecular weight of oxygen (g mol^{-1}).

The volume of the compressed oxygen storage system can be calculated as follows,

$$V_{O_2SYS} = \frac{N_{O_2} \cdot 22.4}{r_{\text{vol}} \cdot P_{\text{tank}}} \left(\frac{T_{\text{op}}}{T_{\text{std}}} \right) 10^{-4} \quad (21)$$

where r_{vol} is the volumetric fraction and P_{tank} is the oxygen tank pressure in MPa, T_{op} is the operating temperature and T_{std} is standard temperature. For present analysis the influence of operating temperature is not considered.

The volume of the liquid oxygen storage system can be calculated as follows,

$$V_{LO_2SYS} = \frac{N_{O_2} \cdot M_{O_2}}{1000 \cdot r_{\text{vol}} \cdot \rho_{LO_2}} \quad (22)$$

where ρ_{LO_2} is the density of liquid oxygen in kg m^{-3} .

3.5. Total volume and weight analysis

Total volume of fuel and oxidant system is estimated based on the available volume after replacing the conventional DG-battery system which is given by,

$$V_{\text{fuelsys}} = \frac{V_{H_2SYS} (V_{\text{DGBAT}} - V_{\text{FC}})}{(V_{H_2SYS} + V_{O_2/LO_2SYS})} \quad (23)$$

and

$$V_{\text{oxisys}} = \frac{V_{O_2SYS} (V_{\text{DGBAT}} - V_{\text{FC}})}{(V_{H_2SYS} + V_{O_2/LO_2SYS})} \quad (24)$$

where V_{DGBAT} is the available volume in m^3 after replacing conventional system and V_{FC} is the volume in m^3 occupied by the fuel cell stacks which is given by,

$$V_{\text{FC}} = \frac{P_{\text{FC}}}{p_{\text{vol_FC}}} \quad (25)$$

where $p_{\text{vol_FC}}$ is the volumetric power density of fuel cell stacks in kW m^{-3} and P_{FC} is the size of the fuel cell stack in kW.

Similarly, total weight of fuel and oxidant system which can be fitted in the submarine after replacing the conventional system can be calculated from the following expressions.

$$W_{\text{fuelsys}} = \frac{W_{H_2SYS} (W_{\text{DGBAT}} - W_{\text{FC}})}{(W_{H_2SYS} + W_{O_2/LO_2SYS})} \quad (26)$$

and

$$W_{\text{oxisys}} = \frac{W_{O_2SYS} (W_{\text{DGBAT}} - W_{\text{FC}})}{(W_{H_2SYS} + W_{O_2/LO_2SYS})} \quad (27)$$

where W_{DGBAT} is the available weight after replacing conventional system and W_{FC} is the weight of the fuel cell stacks which is given by,

$$W_{\text{FC}} = \frac{P_{\text{FC}}}{p_{\text{gra_FC}}} \quad (28)$$

where $p_{\text{gra_FC}}$ is the gravimetric power density of fuel cell stacks in kW kg^{-1} .

3.6. Submerged endurance

A very important factor for the design of submarine is the submerged endurance, which refers to the duration for which the submarine can sail without accessing atmospheric air and it will depend on the total energy which can be stored in the submarine. The submerged endurance based on available weight and volume in days can be estimated from the following equations:

$$E_{\text{store}}^W = \frac{(W_{\text{DGBAT}} - W_{\text{FC}})}{(W_{H_2SYS/SBH} + W_{O_2/LO_2SYS})} \cdot \frac{E_T}{\eta_{\text{OVERALL}}} \quad (29)$$

and

$$E_{\text{store}}^V = \frac{(V_{\text{DGBAT}} - V_{\text{FC}})}{(V_{H_2SYS/SBH} + V_{O_2/LO_2SYS})} \cdot \frac{E_T}{\eta_{\text{OVERALL}}} \quad (30)$$

For present estimation, the lower value between the stored energy based on available weight and volume is considered. As a result of that, the difference in the energy may cause buoyancy matching problem or excess volume.

However, the submerged time also depends on the daily energy requirement which includes the hotel load and propulsion load. The endurance based on daily load profile which includes hotel load and constant propulsion load per day can be estimated taking weight and volume as constraints as follows:

$$T_{\text{end}}^W = \frac{(W_{\text{DGBAT}} - W_{\text{FC}})}{(W_{H_2SYS/SBH} + W_{O_2/LO_2SYS})} \quad (31)$$

and

$$T_{\text{end}}^V = \frac{(V_{\text{DGBAT}} - V_{\text{FC}})}{(V_{H_2SYS/SBH} + V_{O_2/LO_2SYS})} \quad (32)$$

Though the hotel loads can be considered as constant, the propulsion loads requirements may vary. Hence, the total daily energy

requirements may vary based on the propulsion requirements. Hence, the total endurance as a function of propulsion load considering volume as constrain is given by,

$$T_{\text{end}}^V = \frac{E_{\text{store}}^V \cdot \eta_{\text{OVERALL}} - (E_{\text{hor}} + P_{\text{eco}} \cdot T_{\text{eco}} + P_{\text{sprint}} \cdot T_{\text{sprint}})}{E_{\text{hotel}}} \quad (33)$$

$$\text{for } (T_{\text{eco}} + T_{\text{sprint}}) \leq 24 \cdot T_{\text{end}}^V$$

where E_{hor} is the energy required for leaving and entering harbour, P_{eco} is the propulsion power required for sailing, P_{sprint} is the propulsion power required for sprinting, E_{hotel} is the daily hotel load, T_{eco} is the sailing duration, T_{sprint} is the sprinting duration.

Similarly, considering weight as constrain the endurance can be estimated as

$$T_{\text{end}}^W = \frac{E_{\text{store}}^W \cdot \eta_{\text{OVERALL}} - (E_{\text{hor}} + P_{\text{eco}} \cdot T_{\text{eco}} + P_{\text{sprint}} \cdot T_{\text{sprint}})}{E_{\text{hotel}}} \quad (34)$$

$$\text{for } (T_{\text{eco}} + T_{\text{sprint}}) \leq 24 \cdot T_{\text{end}}^W$$

and the total distance which can be travelled by the submarine is given by,

$$S = T_{\text{eco}} \cdot v_{\text{eco}} + T_{\text{sprint}} \cdot v_{\text{sprint}} \quad (35)$$

where v_{eco} is the propulsion power at low speed and v_{sprint} is the propulsion power at high speed.

4. Results and discussions

In the present analysis, metal hydride and sodium borohydride are being considered for hydrogen storage. Hence, the volumetric and gravimetric energy densities are used to estimate the total volume and weight of hydrogen storage system. Additionally a balance of plant (BoP) percentage of weight and volume can be added to ascertain the parameters for the full system based on sodium borohydride as hydrogen carrier. For weight and volume calculations power to weight ratio, density are taken from the data summarised in Table 2.

The value of gravimetric and volumetric ratio for cryogenic oxygen storage varies with the size of the storage tank. For the present calculation volumetric and gravimetric fractions corresponding to 10 m³ tank sizes are adopted.

4.1. Component sizing

The three different configurations which are being considered for analyses are only fuel cell, only battery and battery-fuel cell hybrid system. The daily energy demand for the system is considered constant based on the profile shown in Fig. 2 where, the sprinting speed of the motor is considered as 16 knots. In the fuel cell based system the total energy is fuelled by the fuel cell directly. In hybrid system, during sprinting it is supplied by the battery which is being charged in remaining 22 h of the day to reduce the fuel cell peak power.

4.1.1. Battery bank

The battery capacity is estimated using Equation (4) for various discharge durations. The energy required from the battery and fuel cells for all systems is summarised in Table 3.

4.1.2. Fuel and oxidant system

The weight of hydrogen required for various systems is as calculated from Equation (9). The amount of hydrogen required to meet the load depends upon the efficiencies of other components. The efficiency of different components has been assumed to be

Table 2

Parameters for different components in different systems.

System component	Parameter	Value
Submarine	Total displacement (kg)	3,000,000
Power and energy storage	Total volume (m ³)	327
	Total weight (kg)	406,000
Fuel cell [16]	Gravimetric power density (kW kg ⁻¹)	0.3
	Volumetric power density (kW m ⁻³)	290
	Efficiency	
	Part load (only hotel load)	0.55
	Full load (hotel + motor)	0.44
Storage Battery	Volume (m ³)	0.281
	Weight (kg)	800
	Nominal discharge duration (h)	20
	Efficiency (%)	90
	Depth of discharge	0.5
Metal hydride (MH)	Higher heating value of H ₂ (kW h mol ⁻¹)	0.07944
	H ₂ storage (wt%)	2.25
	System weight (%)	20
	Gravimetric storage capacity (kW h kg ⁻¹)	0.75
	Volumetric storage capacity (kW h m ⁻³)	1200
Sodium borohydride (SBH) [28]	Density (kg m ⁻³)	1074
	Molecular weight (g mol ⁻¹)	37.83
	H ₂ (wt%)	10.66
Compressed oxygen [35]	Storage pressure (MPa)	70
	Weight of empty cylinder (kg)	117.3
	Oxygen weight (kg)	43.2
	Gravimetric fraction	0.265
	Volumetric fraction	0.2
Liquid oxygen (LOX) [36]	Volume of oxygen per tank (m ³)	10
	Density (kg m ⁻³)	1141
	Gravimetric fraction	0.61
	Volumetric fraction	0.35
DC/DC converter	Efficiency (%)	90
Balance of plant	Volume of BoP (%)	10
	Weight of BoP (%)	10
Overall	Efficiency (%)	44
	Energy for leaving and entering harbour (kW h)	250

constant including fuel cell as most of the time it operates at constant load. However, the efficiency of fuel cell changes with the change in load. Hence, in case of only fuel cell system as the fuel cell size is very high, there will be a substantial difference in efficiencies between the part load (22 h) and full load (2 h). For the present analysis, the efficiency of the fuel cell is considered to be 44%.

Table 3

Battery and fuel cell sizes based on daily load.

Configurations	Only FC	Hybrid	Only BAT
Daily energy demand (kW h)	12,222	12,222	12,222
Battery operating duration (h)	0	2	24
Fuel cell operating duration (h)	24	22	0
Energy demand fulfilled by			
Battery (%)	0	39	100
Fuel cell (%)	100	61	0
Energy demand fulfilled by battery (kW h)	0	6449	13,580
Battery size with 50% DOD	0	12,899	27,160
Peak load during fuel cell operation (kW)	2902	475	0
Daily average load (kW)	509	650	0
FC op. load (kW)	2902	650	0
FC size (kW)	3225	650	0

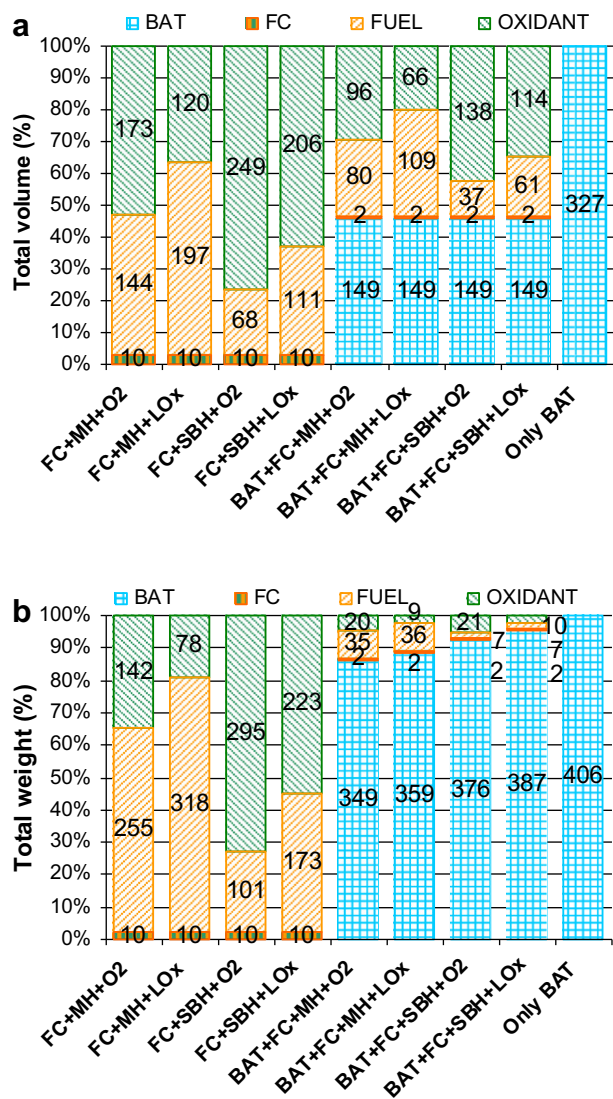


Fig. 5. Analysis of the (a) volume and (b) weight of different components for different systems.

As all technical issues pertaining to the usage of sodium borohydride as hydrogen carrier are not established, the weight and volume of the balance of plant is assumed to be 10% in addition to water required for releasing hydrogen as given in Equation (12). The weight

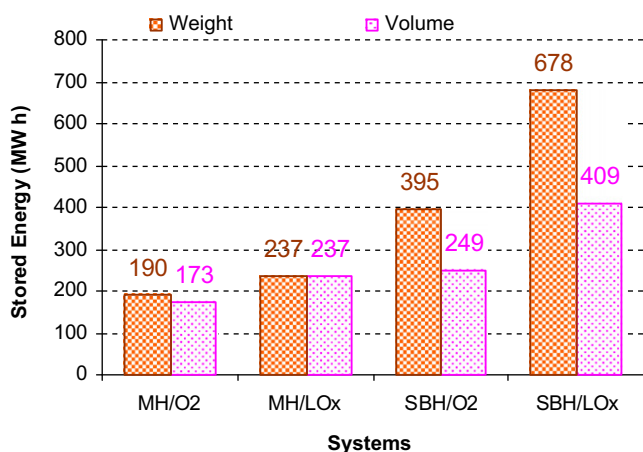


Fig. 6. Comparison of different systems based on the stored energy.

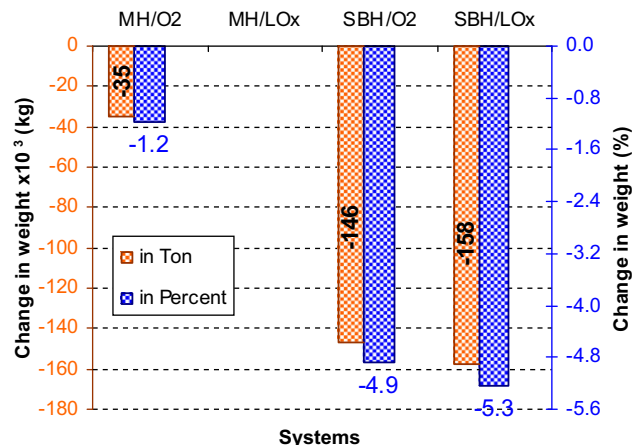


Fig. 7. Total deficit of weight for meeting daily load.

of oxygen required is dependent upon the hydrogen consumption. Hence, the required oxygen is calculated based on hydrogen.

4.2. System comparisons

For retrofitting of submarines, the most important factors are the weight and volume. Hence, in the present analysis volume and weight is considered as constrain for the retrofitting of the 3000 T submarine with AIP system. All the three systems are initially analysed to find out the best system in terms of total fuel, oxidant fuel cell or/and battery can be carried as they determine the total submerged endurance. The analyses of the different systems based on the volume are given in Fig. 5a. For fuel cell based systems all four combinations of fuel and oxidant are considered. From Fig. 5a it is clear that in only fuel cell based system fuel cell occupies only about 3% of the total volume. Hence, the remaining volume could be used for carrying fuel and oxidant. However, in the fuel cell-battery-based hybrid system almost 45% of the total volume is occupied by the battery, which contributes in reducing the volume occupied by the fuel cells only by 8%. Hence, there is an overall loss in the total volume of 37% compared to the only fuel cell based system which will affect the submerged endurance. The analyses of the different systems based on the weight show that the total weight required for the battery in hybrid system and only battery-based system is 424 Mg and 575 Mg respectively, which are much higher than the available weight, 406 Mg. However, in the case of only fuel cell system, fuel cell has only 3% of the total available

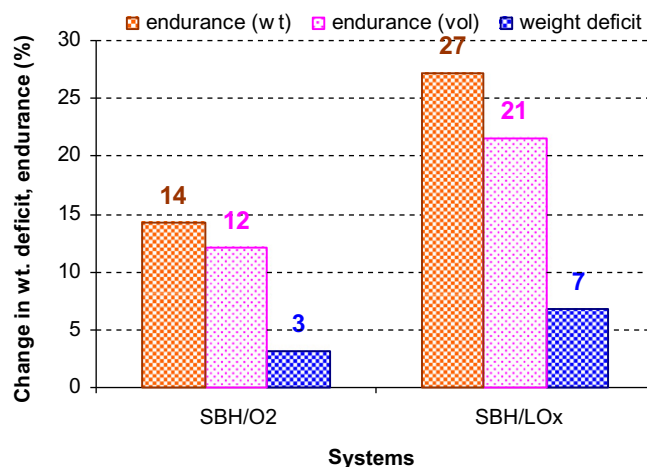


Fig. 8. Change in weight deficit and endurance without water for SBH.

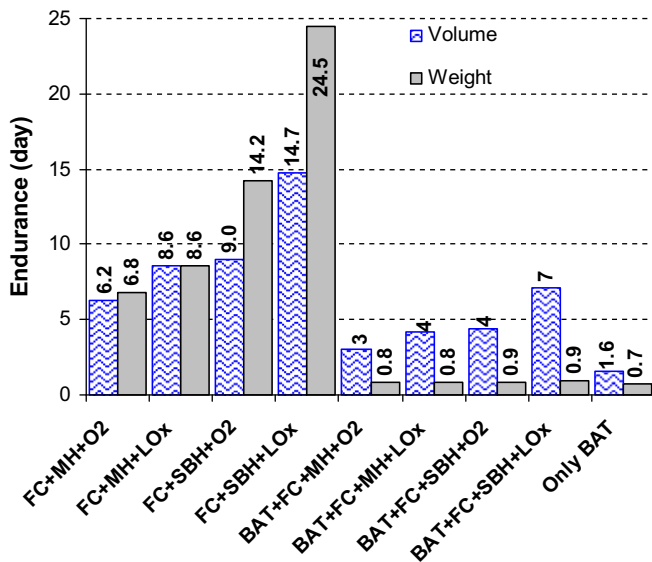


Fig. 9. Analysis of the endurance for different systems considering (a) volume and (b) weight as constrain.

weight leaving a huge weight to carry fuel and oxidant. Hence, it is clear that both the systems are not capable of meeting the daily load considered for the present analysis. With the reduced daily demand for hybrid system and only battery-based system the weight share by different components is presented in Fig. 5b.

From Fig. 5a and b, it is clear that the only fuel cell based system shows favourable results. The hybrid system with batteries for two hours of the load requires very high battery size and thus adversely affects the overall weight and volume.

Total amount of energy that could be stored in the available space is very important for the submerged duration of the submarine. The comparison of total energy that can be stored based on weight and volume is given in Fig. 6. The system with SBH/LO_x is capable of storing maximum energy. However, the analysis based on weight and volume differs substantially. Minimum values between the results obtained based on volume and weight are considered as actual stored energy in retrofitted system. As all the configurations except MH/LO_x are constrained by the volume, the whole system will be facing deficit in total weight.

The analyses of the total deficit in weight for the all the configurations of fuels and oxidants are shown in Fig. 7. On the primary Y-

axis, the total deficit and on the secondary Y-axis, the percentages of deficit based on total weight of the submarine (3000 Mg) for different configurations are shown. From Fig. 7, it is clear that though the sodium borohydride based system shows higher endurance, it faces overall system weight deficit. Hence, the MH/LO_x configuration is best for retrofitting. However, for other configurations it is important to overcome the volume constrain by creating extra volume to compensate the weight deficit by additional fuel and oxidant. The total additional volume required for MH/O₂, SBH/O₂ and SBH/LO_x is respectively 32 m³, 186 m³, and 209 m³.

In the present calculation for SBH based system, it is assumed that the total amount of water required to extract hydrogen from SBH is carried. However, it is also possible to use the water produced from fuel cell as the total water produced by the fuel cell is double the amount required for hydrogen generation. Hence, if the weight of water is excluded in the analysis the endurance is enhanced by 14% and 27% for compressed oxygen and liquid oxygen respectively as shown in Fig. 8. Similarly, if the volume of the water is excluded in the analysis the endurance is enhanced by 12% and 21% respectively for compressed oxygen and liquid oxygen. However, the weight of water has to be replaced by fuel and oxidant and the excess volumes required for SBH/O₂ and SBH/LO_x are respectively, 196 m³ and 233 m³ instead of 186 m³ and 209 m³.

The analysis of the different system considering the total available volume and weight as constrain is shown in Fig. 9a and b. From Fig. 9, it is clear that all the systems are capable of storing less energy compare to the analysis obtained from volume constrain. It is also clear from the endurance analysis that for only fuel cell based system retrofitting will be constrained by the volume for all the fuels and oxidant storage option. Thus, the system will be facing deficit of weight while retrofitting the system which is important for the submarine as it may influence the overall buoyancy of the system.

4.3. Submerged endurance

Submerged endurance is the one of the most important parameter for submarine. The total endurance of the different systems based on the weight volume and weight corresponding to Figs. 5a and 4b is shown in Fig. 9. The poor performances of the battery-fuel cell based hybrid system and only battery system as shown in Fig. 5 are reflected in the submerged endurances. The incapability to meet the present daily load calculated based on the volume for the hybrid and only battery system is also reflected in the endurance as the endurance becomes less than one day. As weight and volume matches for the MH/LO_x configuration, the endurance is found to be

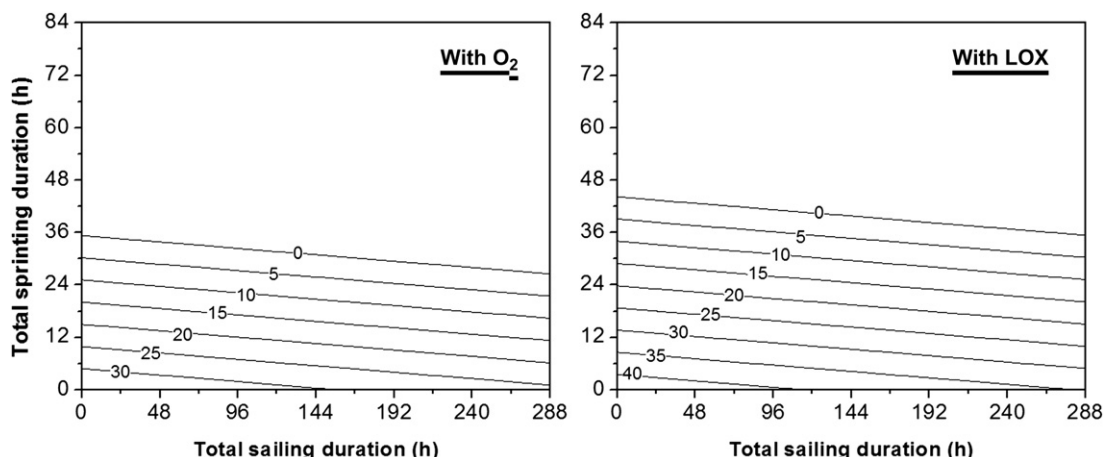


Fig. 10. Variation of endurances (days) with propulsion duration for different hotel loads with MH storage for 100 kW hotel load.

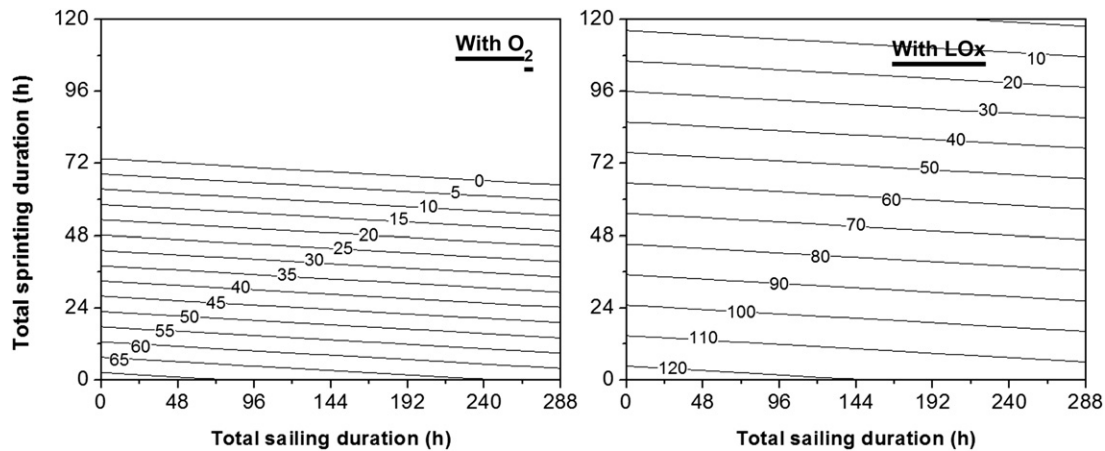


Fig. 11. Variation of endurance (days) with propulsion duration for different hotel loads with SBH storage for 100 kW hotel load.

9 days. However, the analyses based on weight show higher endurance for the MH/O₂, SBH/O₂ and SBH/LO_x. However, this could be achieved only with the introduction of additional volume as already mentioned. Hence, MH/O₂, SBH/O₂ and SBH/LO_x provide at a stretch submerged endurance respectively 7 days, 14 days and 24 days. In contrast the conventional system can provide for an approximate submerged duration of 20 h. Though the load that the present study assumes has almost three times compared to the conventional system it shows higher endurance.

As the propulsion load may change based on the requirement of sprinting (16 knot) or sailing (5 knot) it is more logical to take propulsion load as variable and hotel load as constant as it remains merely constant under submerged condition. The variations in the submerged endurance as a function of propulsion duration are given in Figs. 10 and 11 for hotel loads equal to 100 kW. For this it is assumed that MH/LO_x system will be retrofitted and for rest of the three systems additional volume will be introduced to overcome the volume constrain in the system. The endurance is reduced linearly with the increase in the hotel load. The total distances covered by the submarine increase with the sprinting and sailing durations.

4.4. Limiting curve

From Figs. 10 and 11, it is clear that with the increase in sailing and sprinting duration the submerged endurance decreases. Hence, there will be situation when the propulsion duration will be equal to the total endurance. Beyond this, the total submerged time becomes less than the total propulsion duration. Hence, the limiting curves put the restriction on maximum propulsion duration. Any point on the limiting curve represents the minimum submerged endurance and maximum distance could be travelled by the submarine. The upper side of the limiting curves represents infeasible regions as in this region the submerged duration becomes lower than the total propulsion duration. Hence, the limiting curve on the endurance plot gives the minimum endurance if combined with endurance plot and on the distance plot it gives the maximum covered distance.

The limiting curve obtained for different MH and SBH based systems at three different average hotel loads is shown in Fig. 12. It is clear from Fig. 12 that the SBH based system with LO_x has minimum infeasible region whereas MH based system with compressed oxygen offers minimum feasible region. Hence, the maximum distance could be covered during submerged condition

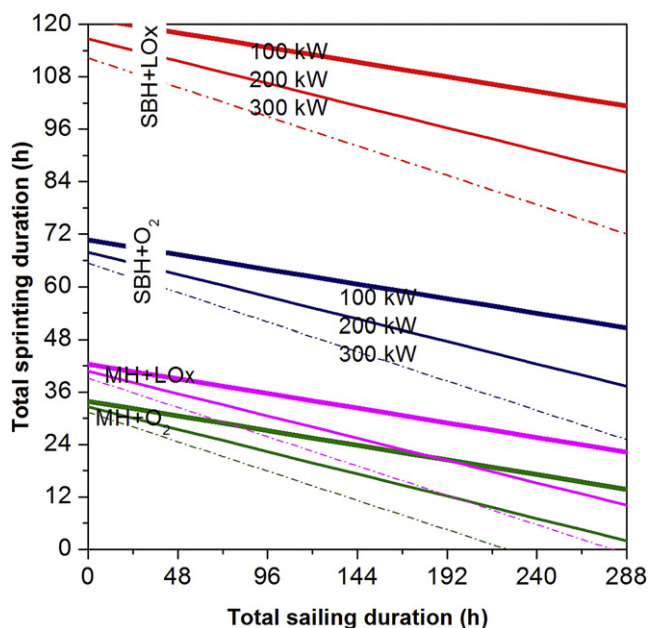


Fig. 12. Boundary lines for different hotel loads (a) MH and (b) SBH with water based system.

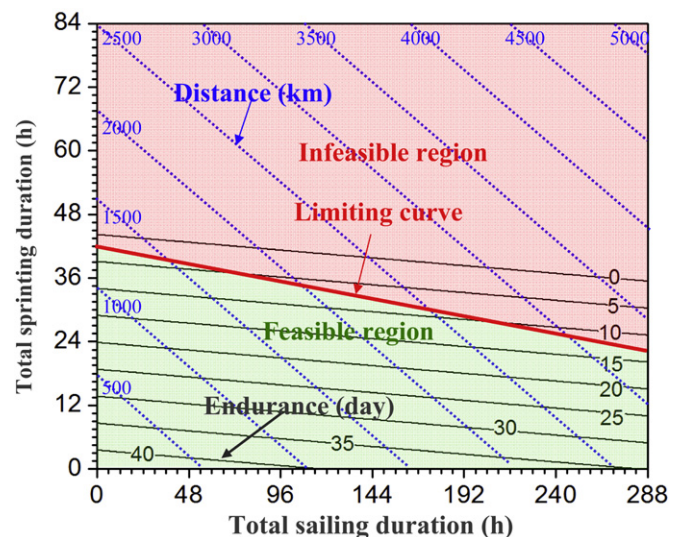


Fig. 13. Combined plot of limiting curve, submerged endurance and total distance for MH + LO_x system with average hotel load equals to 100 kW.

is maximum for SBH + LO_x system and for MH + O₂ system it is minimum as the distance increases with the propulsion duration. However, to analyse the submerged endurance the limiting curve must be combined with the endurance plot. One such combined plot for MH + LO_x system and average load equal to 100 kW is plotted in Fig. 13. It could be very useful to analyse the total distance and submerged endurance for any given average hotel load. Say for example, if it is planned for 18 h of sailing and 28 h of sprinting it will give 15 days of endurance and total 1000 km could be travelled under submerged condition and as the corresponding point is far away from the limiting curve the total propulsion duration is much less than the submerged duration. However, for the case of 180 h and 28 h of sailing and sprinting duration respectively, the submerged endurance reduced to 10 days and total 2500 km could be travelled. Whereas, in such case the total propulsion duration becomes almost equal to the submerged endurance as the point on the combined plot comes close to the limiting curve.

5. Conclusions

Various possible combinations of fuel cell and battery are analysed for air independent propulsion based submarine. In this purpose the weight, volume and efficiency are taken into considerations. The weight and volume analyses favour the only fuel cell based system. Among different combinations for fuel and oxidant, it is found that the MH/LO_x system could be simply retrofitted with fuel cell system. However, for MH/O₂, SBH/O₂ and SBH/LO_x an extra room has to be created to compensate the deficit of volume respectively 32 m³, 186 m³ and 209 m³. It is observed that the total submerged endurance is enhanced substantially with fuel cell system. With SBH and LO_x the minimum submerged endurance could be 6.5 days with almost 120 h of sprinting with a total travelled of 3500 km. However, the submerged endurance is improved to 33 days if the sprinting time is reduced to 84 h and an introduction of sailing time of 90 h with same distance to be travelled.

References

- [1] Adams VW. Possible fuel cell applications for ships and submarines. *J Power Sources* 1990;29:181–92.
- [2] Sattler G. PEFC's for naval ships and submarines: many tasks, one solution. *J Power Sources* 1998;71:144–9.
- [3] Leo TJ, Durango JA, Navarro E. Exergy analysis of PEM fuel cells for marine applications. *Energy* 2010;35:1164–71.
- [4] Psoma A, Sattler G. Fuel cell systems for submarines: from the first idea to serial production. *J Power Sources* 2002;106:381–3.
- [5] Brighton DR, Mart PL, Clark GA, Rowan MJM. The use of fuel cells to enhance the underwater performance of conventional diesel electric submarines. *J Power Sources* 1994;51:375–89.
- [6] Saxe M, Folkesson A, Alvfors P. Energy system analysis of the fuel cell buses operated in the project: clean urban transport for Europe. *Energy* 2008;33:689–711.
- [7] Appleby AJ. Fuel cell technology: status and future prospects. *Energy* 1996;21:521–653.
- [8] Berry GD, Pasternak AD, Rambach GD, Smith JR, Schock RN. Hydrogen as a future transportation fuel. *Energy* 1996;21:289–303.
- [9] Sharifi Asl SM, Rowshanzamir S, Eikani MH. Modelling and simulation of the steady-state and dynamic behaviour of a PEM fuel cell. *Energy* 2010;35:1633–46.
- [10] Thampan T, Malhotra S, Zhang J, Dutta R. PEM fuel cell as a membrane reactor. *Catal Today* 2001;67:15–32.
- [11] Brown LF. A comparative study of fuel cells for onboard hydrogen production for fuel cell powered automobiles. *Int J Hydrogen Energy* 2001;26:381–97.
- [12] Strasser K. Mobile fuel cell development at Siemens. *J Power Sources* 1992;37:209–19.
- [13] San Martin JL, Zamora I, San Martin JJ, Aperribay V, Torres E, Eguia P. Influence of the rated power in the performance of different proton exchange membrane (PEM) fuel cells. *Energy* 2010;35:1898–907.
- [14] Gencoglu MT, Ural Z. Design of PEM fuel cell system for residential application. *Int J Hydrogen Energy* 2009;34:5242–8.
- [15] Lipman TE, Edwards JL, Kammen DM. Fuel cell system economics: comparing the costs of generating power with stationary and motor vehicle PEM fuel cell systems. *Energy Policy* 2004;32:101–25.
- [16] Hissel D, Péra MC, Kauffmann JM. Diagnosis of automotive fuel cell power generators. *J Power Sources* 2004;128:239–46.
- [17] Rodatz P, Büchi F, Onder C, Guzzella L. Operational aspects of a large PEFC stack under practical conditions. *J Power Sources* 2004;128:208–17.
- [18] Wang L, Husar A, Zhou T, Liu H. A parametric study of PEM fuel cell performances. *Int J Hydrogen Energy* 2003;28:1263–72.
- [19] Balat M. Potential importance of hydrogen as a future solution to environmental and transportation problems. *Int J Hydrogen Energy* 2008;33:4013–29.
- [20] Montignac F, Noirot I, Chaudourne S. Multi criteria evolution of on board hydrogen storage technologies using the MACBETH approach. *Int J Hydrogen Energy* 2009;34:4561–8.
- [21] Weinberger B, Lamari FD. High pressure cryo storage of hydrogen by adsorption at 77 K and up to 50 MPa. *Int J Hydrogen Energy* 2009;34:3058–64.
- [22] Sakintuna B, Lamari-Darkrim F, Hircher M. Metal hydride materials for solid hydrogen storage: a review. *Int J Hydrogen Energy* 2007;32:1121–40.
- [23] Darkrim FL, Malbrunot P, Tartaglia GP. Review of hydrogen storage by adsorption in carbon nanotubes. *Int J Hydrogen Energy* 2002;27:193–202.
- [24] Mori D, Hirose K. Recent challenges of hydrogen storage technologies for fuel cell vehicles. *Int J Hydrogen Energy* 2009;34:4569–74.
- [25] Sarkar A, Banerjee R. Net energy analysis of hydrogen storage options. *Int J Hydrogen Energy* 2005;30:867–77.
- [26] Ahluwalia RK, Hua TQ, Peng J-K. Fuel cell efficiencies of different automotive on-board hydrogen storage options. *Int J Hydrogen Energy* 2007;32:3592–602.
- [27] Çakanyıldırım Ç, Gürü M. Hydrogen cycle with sodium borohydride. *Int J Hydrogen Energy* 2008;33:4634–9.
- [28] Umegaki T, Yan JM, Zhang XB, Shioyama H, Kuriyama N, Xu Q. Boron- and nitrogen-based chemical hydrogen storage materials. *Int J Hydrogen Energy* 2009;34:2303–11.
- [29] Liu BH, Li ZP. A review: hydrogen generation from borohydride hydrolysis reaction. *J Power Sources* 2009;187:527–34.
- [30] Gislón P, Monteleone G, Prosini PP. Hydrogen production from sodium borohydride. *Int J Hydrogen Energy* 2009;34:929–37.
- [31] Minkina VG, Shabunya SI, Kalinin VI, Martynenko VV, Smirnova AL. Long term stability of sodium borohydrides for hydrogen generation. *Int J Hydrogen Energy* 2008;33:5629–35.
- [32] Fakioglu E, Yürüm Y, Veziroglu TN. A review of hydrogen storage systems based on boron and its compounds. *Int J Hydrogen Energy* 2004;29:1371–6.
- [33] Lia ZP, Liu BH, Arai K, Suda S. Development of the direct borohydride fuel cell. *J Alloy Compd* 2005;404–406:648–52.
- [34] Twidell JW, Weir AD. Renewable energy resources. 1st ed. NY: E & F.N. Spon Ltd; 1986.
- [35] Storhy final event – Hydrogen storage systems for automotive applications, <http://www.storhy.net> from Internet website www.google.com for 'Hydrogen storage at 700 bar' on 14 December 2008.
- [36] Cryogenic Storage Tanks: Technical Specifications, <http://www.universal-boschi.com/cryogenic-storage-tanks.html> on 22 July 2010.